

## Vision-Based TicTacToe Robot

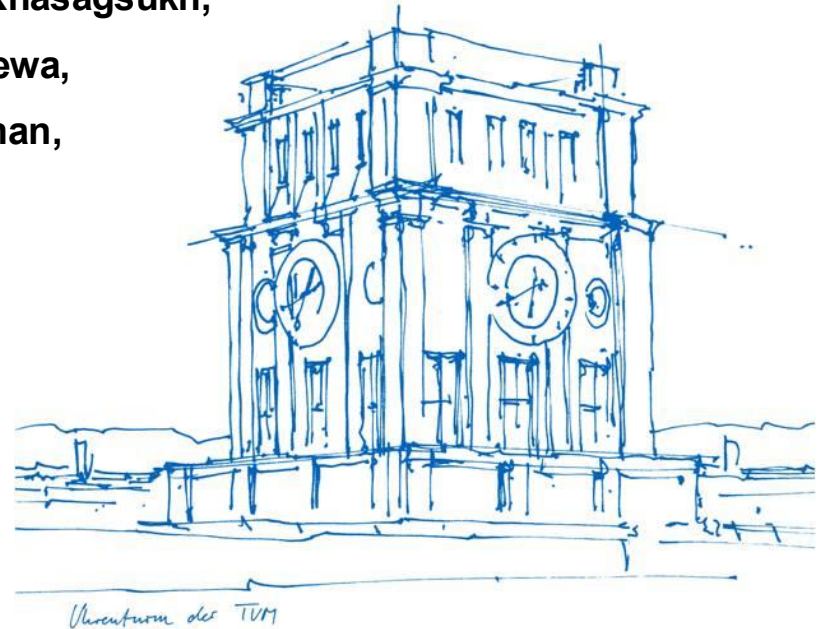
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CPS Project Final Presentation

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# Our CPS Project Team



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# Outline

1. Motivation & Project Goals
2. System Concept & Architecture
3. Hardware Components & Setup
4. Features of TicTacTUM
5. Challenges Faced
6. Improvements & Future Work
7. Conclusion & Q&A

# Motivation & Project Goals

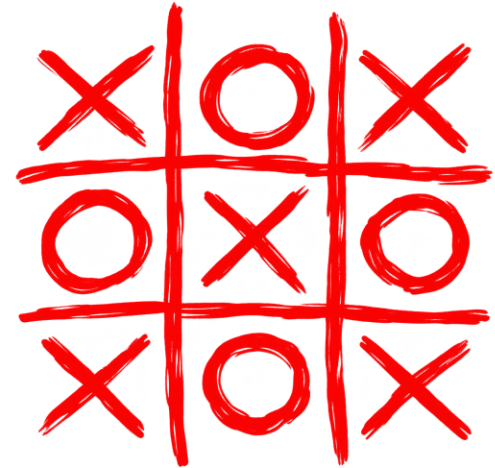
# Motivation & Project Goals

## Motivation

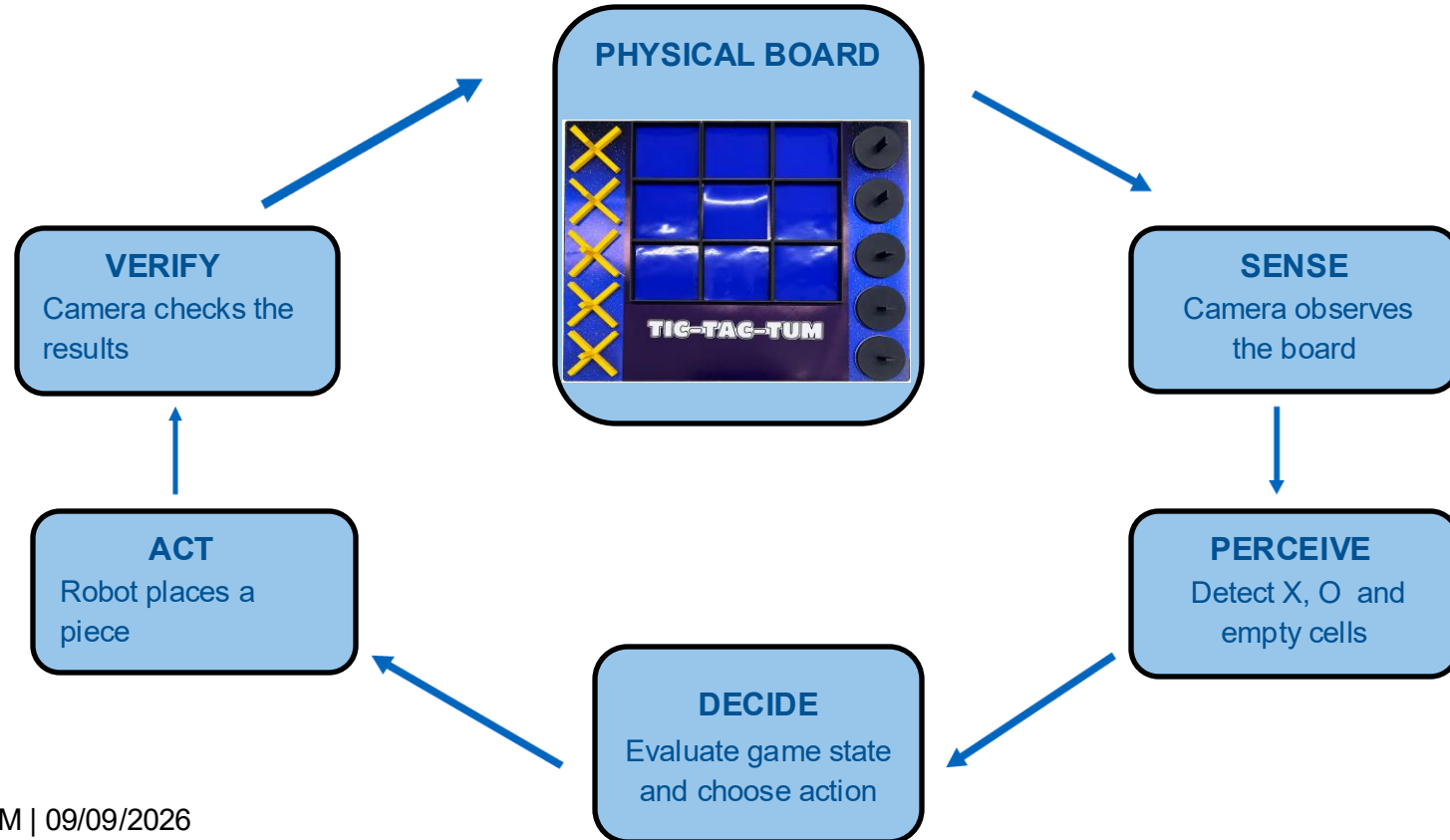
- Traditional Tic-Tac-Toe is usually played in a **virtual environment**. Our project brings the game into the **physical world** by combining computer vision, game logic, and robotic manipulation.

## Goal

- The goal of our project is to develop a system that can **observe** the physical board, **understand** the current game state, **perform** a move using a robotic arm, and **verify** the result using the camera.

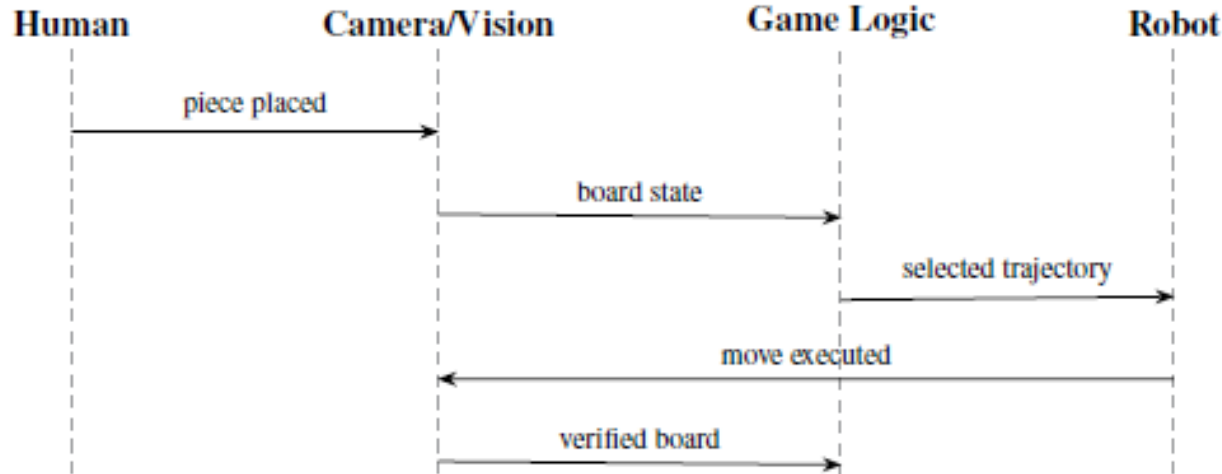


# From Game to Cyber-Physical System



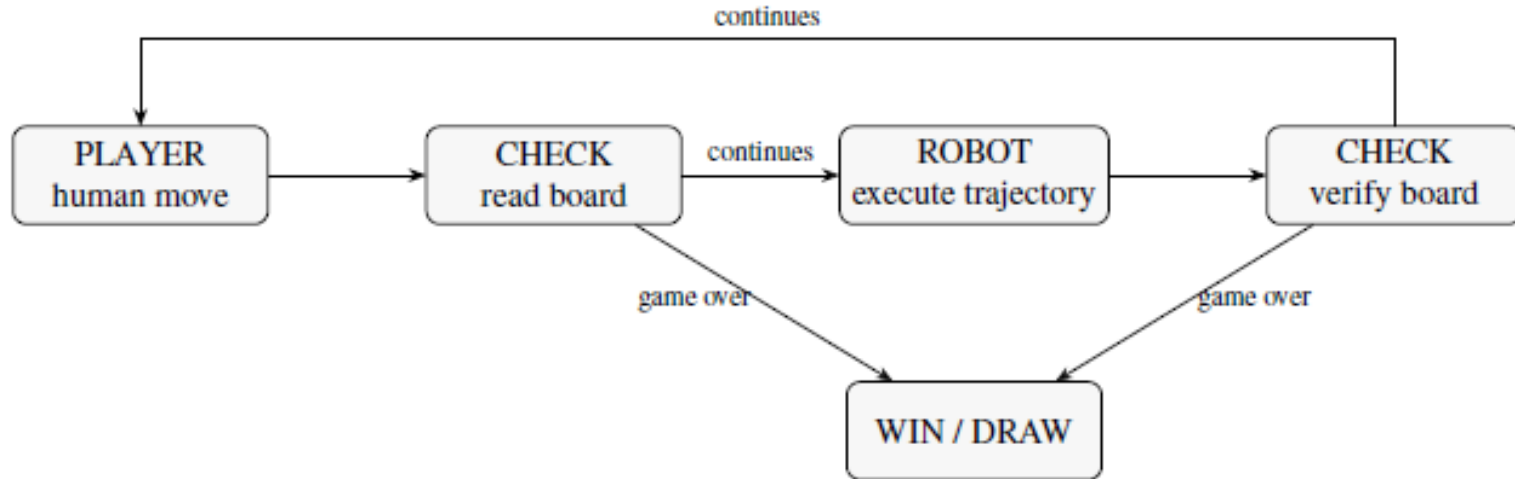
# System Concept & Architecture

# System Architecture & Interaction Flow



- **Perception:** Camera detects the physical board
- **Decision:** Game logic evaluates the board state
- **Action & Verification:** Robot executes the move and the camera verifies it

# Game State & Control Flow



- Each **CHECK** updates the physical board state and determines whether the game continues.

# Hardware Components & Setup

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SO-101 Robot  
Arm



Camera



3 × 3 Game  
Board



Servo Driver  
Board



6 x Feetech  
ST3215 Servo  
Motor

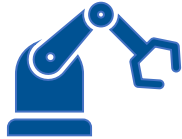


Host Computer



# Features of TicTacTUM

# Features of TicTacTUM (I/II)



**Live board detection** using a camera and OpenCV

**Interactive board calibration** by selecting four board corners

**Fullscreen visual feedback** of the detected board



**Perspective correction** to a normalized  $300 \times 300$  pixel view

**Automatic cell classification** of yellow X, dark/orange-grey O, or empty cells

**Real-time board state recognition** after every move



**Win and draw detection** across rows, columns, and diagonals

**Minimax robot strategy** to select the best available move

**Player-first or robot-first mode**

# Features of TicTacTUM (II/II)



**SO-101 robotic arm control**  
using the LeRobot framework

**Recorded trajectory playback**  
for predefined board positions

**Six-joint trajectory control** for  
arm and gripper movement

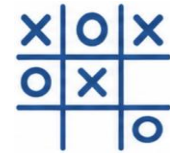
**20 Hz recording and playback**  
for smooth execution



**Replay and recalibration**  
**controls** during the game

**Main menu navigation** for  
restarting or changing settings

**Robot simulation fallback**  
when LeRobot is unavailable



**Fullscreen OpenCV interface**  
adapted to the screen resolution

**Text-to-speech feedback** for  
turns, board checks, and results

**Physical and standalone**  
**gameplay modes** in one  
system

# Challenges Faced

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- Different lighting conditions affected board and piece detection.
- Robot sometimes struggled to pick up pieces correctly.
- The robot needs consistent maintenance for the screws.
- Schedule conflicts to meet up and work together.

# Improvements & Future Work

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- The setup can be improved into playing different board games.
- A second arm can be used to make the arms play against each other.
- The system can be trained to identify object orientation so the arm can pick any object up.

# Conclusion & Q&A

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- TicTacTUM successfully combines computer vision, game logic, and robotic control to create a physical tic-tac-toe game.

